# 4.9 Modeling with Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.9 · Student sheet

**I can…**

- write cubic functions given the x-intercepts and a point.
- use finite differences to find the degree of a polynomial model.
- use technology to find polynomial models for real-life data.

## Part 1 · Stacking cubes

Square pyramids of cubes: 1 layer = 1 cube, 2 layers = 1 + 4 cubes, 3 layers = 1 + 4 + 9 cubes, …

| Layers | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Cubes | | | | | | |
| First differences | | | | | | |
| Second differences | | | | | | |
| Third differences | | | | | | |

Degree of the model: _____

## Part 2 · Notes: fill in the blanks

Cubic from x-intercepts p, q, r: f(x) = _____. Use another point to find _____.

Constant nth differences → degree _____.

## Part 3 · Examples (work them with the class)

### Example 1

Write the cubic function whose graph has x-intercepts −2, 1, and 3 and passes through (0, −6).

### Example 2

Use finite differences to find the degree of the polynomial function that fits the data. Then find the function.

| x | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| f(x) | 2 | 9 | 28 | 65 | 126 | 217 |

### Example 3

Write the cubic function with x-intercepts $-1$, 2, and 4 that passes through (1, 12).

## Part 4 · Partner practice

### Partner 1

Find the degree of the polynomial that fits the data: $x = 1, 2, 3, 4, 5$; $y = 3, 6, 11, 18, 27$.

### Partner 2

Write the cubic function with x-intercepts $-3$, 0, and 2 that passes through $(1, -8)$.